

# Yang Liu

Postdoctoral Researcher, University of Helsinki

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## EDUCATION

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### University of Helsinki

Ph.D. in Computer Science, defended on August 11th (Opponent: Joseph A. Konstan)

Helsinki, Finland

Sep 2020–Aug 2025

- Thesis: “Methodological and Extrinsic Challenges in Offline Evaluation of Recommender Systems”
- Advisors: Dorota Glowacka, Alan Medlar

### University of Helsinki

M.S. in Computer Science, GPA: 5/5

Helsinki, Finland

Sep 2018–Mar 2020

- Thesis: “Assessing Text Readability and Quality with Language Models”
- Advisors: Dorota Glowacka, Alan Medlar

### Xidian University

B.S. in Telecommunications Engineering, GPA: 3.42/4

Xi'an, China

Aug 2013–Jul 2017

- Thesis: “The Influence of Atmospheric Turbulence for Wireless Optical Communication”

## RESEARCH EXPERIENCE

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### Postdoctoral Researcher, University of Helsinki

(Upon Ph.D. completion, expected Sep 2025)

Sep 2025 – Present

### Doctoral Researcher, University of Helsinki

Apr 2020 – Aug 2025

- **Meta-evaluation of recommender systems**
  - \* User perceptions of explanations in cross-domain recommendation ([SIGIR 2024](#))
  - \* Comprehensive meta-evaluation framework of recommender systems ([RecSys 2023](#))
  - \* Evaluation flaws of recommender systems ([RecSys 2023](#), [2025](#))
- **Exploratory search of generative models**
  - \* Intelligent system for personalized user interaction with GANs ([SIGIR 2022](#), [IUI 2024](#))
- **Semantic shifts detection in NLP**
  - \* Lexical ambiguity detection in professional discourse ([IPM 2022](#))
  - \* Robust semantic shift detection using BERT ([Eval4NLP 2022](#))

### Research Assistant, University of Helsinki

Jun 2019 – Apr 2020

- **NLP for text analysis**
  - \* Automatic text quality assessment using various language models ([ICTIR 2021](#))
  - \* Political bias identification of news articles

## PUBLICATIONS

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### Conferences Papers

1. H.L. Huy, **Y. Liu**, A. Medlar, and D. Glowacka, “Don’t Get Ahead of Yourself: A Critical Study on Data Leakage in Offline Evaluation of Sequential Recommenders”, *Proceedings of the 19th ACM Conference on Recommender Systems (RecSys)*, 2025 (to be present)

2. K. Liu, **Y. Liu**, A. Medlar, and D. Glowacka, “Disentangling User and Item Sequence Patterns in Sequential Recommendation Data Sets”, *Proceedings of the 19th ACM Conference on Recommender Systems (RecSys)*, 2025 (to be present)
3. **Y. Liu**, A. Medlar, and D. Glowacka, “Sample, Nudge and Rank: Exploiting Interpretable GAN Controls for Exploratory Search”, *Proceedings of the 29th International Conference on Intelligent User Interfaces (IUI)*, 2024
4. D. Kotkov, A. Medlar, **Y. Liu**, and D. Glowacka, “On the negative perception of cross-domain recommendations and explanations”, *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, 2024
5. **Y. Liu**, A. Medlar, and D. Glowacka, “What We Evaluate When We Evaluate Recommender Systems: Understanding Recommender Systems’ Performance Using Item Response Theory”, *Proceedings of the 17th ACM Conference on Recommender Systems (RecSys)*, 2023
6. **Y. Liu**, A. Medlar, and D. Glowacka, “On the Consistency, Discriminative Power and Robustness of Sampled Metrics in Offline Top-n Recommender System Evaluation”, *Proceedings of the 17th ACM Conference on Recommender Systems (RecSys)*, 2023
7. **Y. Liu**, A. Medlar, and D. Glowacka, “Rogue: A System for Exploratory Search of GANs”, *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, 2022
8. A. Medlar, J. Li, **Y. Liu**, and D. Glowacka, “Critiquing-based Modeling of Subjective Preferences”, *Proceedings of the 30th ACM Conference on User Modeling, Adaptation and Personalization (UMAP)*, 2022
9. **Y. Liu**, A. Medlar, and D. Glowacka, “Can language models identify Wikipedia articles with readability and style issues?”, *Proceedings of the 2021 ACM SIGIR International Conference on Theory of Information Retrieval (ICTIR)*, 2021

### Journal Articles

1. **Y. Liu**, A. Medlar, and D. Glowacka, “Lexical Ambiguity Detection in Professional Discourse”, *Information Processing & Management*, 59.5:103000, 2022

### Workshop & Other Publications

1. **Y. Liu**, A. Medlar, and D. Glowacka, “Statistically Significant Detection of Semantic Shifts using Contextual Word Embeddings”, *Proceedings of the 2nd Workshop on Evaluation and Comparison of NLP Systems (Eval4NLP)*, 2021

## SCHOLARSHIPS AND AWARDS (SINCE 2019)

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|                                                                                   |           |
|-----------------------------------------------------------------------------------|-----------|
| • SIGIR Student Travel Grant, \$1,400 plus registration                           | Jul 2024  |
| • SIGCHI Gary Marsden Travel Awards, \$2,500                                      | Sep 2023  |
| • Researcher of the Year (department of computer science, University of Helsinki) | 2023      |
| • Nokia Scholarship, €5,000                                                       | 2022      |
| • Donasci Travel Grant (attending SIGIR2022 in Madrid, €1,000)                    | 2022      |
| • Second Year Study Grant, (€10,000)                                              | 2019–2020 |

## ACADEMIC SERVICES

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- **PC Member:** IUI (Since 2024)
- **Reviewers for Journals:** Information Processing & Management (Since 2022)
- **Subreviewers:** WWW, SIGIR

## TEACHING AND SUPERVISION

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- **Guest Lecturer** of Information Retrieval (DATA 20021), University of Helsinki  
Online controlled experiment, Recommender Systems 2022–2025
- **Bachelor’s Thesis Supervisor** (TKT20013), University of Helsinki Since 2024